

3D Printing

Printing in three dimensions has applications for everyone, from a foodie to an engineer and from our future scientists to our present-day doctors

Raj Saxena

readersletters@tbinkdigit.com

3D printing is also known as additive manufacturing. It is a process of making solid objects from a digital model. This digital model is made by using a specialised software provided by the manufacturer of the printer or by using a 3D modelling software. Here are some implementations.

Prototyping

Prototyping is used in nearly every form of construction to get hold of the problems that the engineered design may face in the future. It is a fundamental step in the development process of any design that takes a lot of time and energy to produce. With new 3D printers, this cycle of development and testing of prototypes become much shorter resulting into better designs in shorter time spans as all you need to do is make the required design in the 3D modelling software and press print. Some-



Industrial design needs 3D prints

thing that used to take weeks in development will now take not more than a few hours. Plus a modification in the original design doesn't take weeks like it used to in the past, just make changes in the 3D model and press print again!

Airbus plans to come up with planes that use parts printed by a 3D printer. The technology has not reached the required level of complexity, and research is ongoing. But Airbus promises to invest enough money to develop a technology that is safe to print high quality material that can be used in the aviation industry. Not only will this decrease the labour costs, but also reduce the time to deliver finished products by more than half.

One for the expecting parents

Research is on to use a 3D printer to "Print" a model of the foetus. By using a 4D scan to capture the image of the foetus and after some processing in the proprietary software, the printer prints a model of the young one who is still months away from being born. This is a breakthrough technique as it gives the doctors a better view of the baby, so in case the foetus requires surgery, it can be planned so that the doctors don't get any surprises in the middle of the surgery.



Fancy taking your baby prints?

No more closed chemist shops

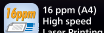
When Professor Lee Cronin of Glasgow University learned what 3D printers could do, he thought of converting it into a drugstore, literally! Professor Lee's research has given meaning to terms such as "downloadable chemistry". The idea (although in its initial stages) is to be able to "print" drugs from the convenience of your home. The team is currently looking at ways in which simple drugs such as Ibuprofen can be successfully produced, using a 3D printer.

"If that principle can be established, then the possibilities suddenly seem endless. Imagine your printer like a refrigerator that is full of all the ingredients you might require to make any dish in Jamie Oliver's new book," Cronin was quoted as saying in British newspaper *The Guardian*. Plans are also in place to make an App



Your one stop solution to printing needs

Space Saving. Power Saving. Easy-to-use. Multi-functional.
The all new range of Printers from Panasonic. It's everything you want it to be.







Log on to www.panasonic.net or email priye.choudhary@in.panasonic.com

*Available in KX-MB1520